



NAKKA HEMANTH DATTA

Engineering Physics | EP23B028

CGPA: 8.45 GitHub LinkedIn

Education

- Indian Institute of Technology Madras** Chennai, TN
B. Tech – Major: Engg. Physics, Minor: AIML Jul '23 – Jul '27
- JNV, Pondicherry** Pondicherry, PY
Class XII (CBSE) | 95.80% 2023

Professional Experience

Wells Fargo | PPO Offered Bangalore, India
Software Engineering Intern | Full Stack & AI Systems May '26 – Jul '26

- Deployed a containerized **3-service trade-processing stack (FastAPI, React, Redis)** on **Kubernetes** with HPA autoscaling & liveness/readiness probes, raising throughput to **10,000+ trade confirmations/hour** & replacing manual review workflows.
- Architected the underlying **extraction engine** using **Strategy Design Patterns, PyMuPDF, and Redis caching**, automating **JSON-to-SQL schema normalization** & sub-second trade database reconciliation.
- Developed a **real-time React review workbench** featuring **SSE-based status streaming**, split-screen verification, embedded PDF inspection, & **pagination** for rapid exception handling.
- Developed a **trade management system** using **Angular 16, ASP.NET Core 8, Dapper, & SQL Server**, optimizing data access via batched SOR retrieval, in-memory caching, & transactional status synchronization.
- Built & tested **LLM-as-a-Judge agent evaluation workflows** (hallucination, truthfulness, response quality) with trace-based observability, using judge-vs-gold calibration to validate scoring reliability for AI applications.

CV. OMSAI INFORMASI TEKNOLOGI Remote, Singapore
Forward Deployed Engineer (Contract) Feb '26 – May '26

- Partnered directly with an **Oman-based enterprise client** to build, iterate, & deploy a production AI document intelligence system on a cloud VM, driving end-to-end solution delivery.
- Engineered a **multimodal Corrective RAG (CRAG)** pipeline using **LangGraph, OpenAI embeddings & LLMs, ChromaDB (vector + BM25), semantic chunking, & a reranking engine** with query-rewrite/re-retrieval on low-confidence hits, achieving **100% table query accuracy across a 250+ enterprise document benchmark**.
- Developed a **React frontend** for source document rendering with **live citation highlighting**, utilizing **Ragas framework context metrics** to trim prompt token usage by **35%** & lower LLM inference latency.

Technical Skills

- Languages:** Python, C/C++, SQL, Matlab, Mathematica
- Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Ragas, FAISS, ChromaDB, Unsloth, FastAPI, React, Angular 16, ASP.NET Core, Dapper, SQL Server, Docker, Kubernetes, Redis, Celery, Streamlit, pandas, Git
- Core Skills:** DSA, Operating Systems, Distributed Systems, Concurrency & Multithreading, SQL Indexing & Query Optimization, Machine Learning, Deep Learning, NLP, Computer Vision, RAG, LLM Fine-Tuning (PEFT/LoRA), LLMops & Observability

Relevant Coursework

- Data Structures & Algorithms
- Database Management Systems
- Probability & Statistics
- Intro to Machine Learning
- Introduction to Deep Learning
- Computer Organisation
- Modern App Development
- Natural Language Processing
- Computer Vision
- Computational Physics

Position of Responsibility

CFI: Sahaay Social Innovation Club
Project Member (Gesturize) Apr '24 – Mar '25

- Co-invented an Indian Sign Language (ISL) translation system on low edge devices using **Siamese Networks, MediaPipe, & interpolation (Patent App. No: 202541022908)**, leading 60+ **NSS IIT Madras** volunteers for open-source dataset contribution.

Mentorship

Avanti Fellows & EE Intern Buddy Program
Student Mentor Sep '23 – Aug '26

- Mentored underprivileged JEE aspirants by giving back as an Avanti Fellows alumnus & guided junior EE students as Mentor.

Projects

Agentic Chat Application – “Hemnn AI (AI Twin)” Jan '26 – Present

- Built an **agentic AI portfolio** using **LangGraph** with intent routing, conversational memory, RAG, & tool calling for grounded access to personal projects, experience, skills, & technical background.
- Engineered a **dynamic multi-agent orchestration layer** routing requests to specialized sub-agents, supporting creation of task-specific agents & enabling plug & play capabilities through **MCP servers**.
- Architected a **privacy-aware backend** deployed on **Azure** with a **Vercel-hosted frontend**, featuring isolated sessions, thread-scoped state, bounded execution.

Colab Sync Control – Open-Source Developer Tool Jul '26 – Present

- Built & published VS Code extension enabling **bi-directional synchronization between local workspaces & remote Google Colab runtimes**, with automatic background syncing & conflict monitoring.
- Architected a **local daemon-based control plane** for workspace linking, remote session lifecycle management, & interactive shell access, integrating cloud compute into the native VS Code workflow.
- Developed a real-time dashboard for **CPU/RAM/GPU compute-unit tracking**, supporting T4/L4/A100 GPU & TPU runtimes.

Distributed High-Performance N-Body Simulation Engine Jan '26 – May '26

- Engineered a **gravitational N-body simulator**, layering naive $\mathcal{O}(N^2) \rightarrow$ vectorized NumPy $\rightarrow$ Barnes-Hut ($\mathcal{O}(N \log N)$) $\rightarrow$ multiprocessing $\rightarrow$ socket-based distributed execution (rel. error $< 10^{-12}$).
- Vectorized force kernel (NumPy) for $\sim 78\times$ **speedup** over naive loop ($N = 256$) & parallelized Barnes-Hut tree traversal across 8 workers for $\sim 3.0\times$ **further speedup** ($N = 8192$).
- Built a coordinator/worker raw-byte protocol ($< 0.1\%$ overhead) with **fail-stop fault tolerance** (checkpoints) & benchmarked 6 scaling experiments across N , speedup, & opening angle θ tradeoffs.

Foundational Model Adaptation Pipeline Jul '25 – Dec '25

- Built a modular **model adaptation pipeline** for CLIP & DINOv2 supporting zero-shot, few-shot, & PEFT workflows.
- Implemented **LoRA, BitFit, & Prefix Tuning** & optimized training to update only **0.7% of model parameters**, achieving **94.67% accuracy** with DINOv2.
- Designed a **two-stage training architecture** separating feature extraction & classifier optimization, enabling efficient experimentation across model & adaptation configurations.

Score-Based Generative Model Evaluation Framework Jul '25 – Dec '25

- Implemented **DDPM, DDPM++, & NCSN++** training & sampling pipelines across VP, VE, & Sub-VP SDE formulations using PyTorch.
- Built a configurable **model evaluation pipeline** supporting 12 architectures, hyperparameter sweeps, class-specific training, & large-scale training runs up to **200K steps**.
- Benchmarked model configurations using **FID & Inception Score**, achieving **FID 10.52 & IS 7.37** with VP-DDPM++ & identifying architecture choices affecting training efficiency & generation quality.

Hackathons

2nd Prize, Mando AI Tool Dev Hackathon 2025 May '25 – Jun '25

- Engineered a full-stack document QA system supporting PDF, DOCX, & CSV, using LangGraph, VoyageAI embeddings, Claude LLM, & FAISS hybrid search across **2,000+ page documents**.

Round 3 Finalist, Appian Algnite Hackathon 2025 Mar '25 – Apr '25

- Built a multimodal product discovery platform using CLIP+FAISS hybrid search across **14K+ products**, with a Gemini-powered AI browser extension for automated e-commerce checkout.

Scholastic Achievements

- CBSE Certificate of Merit for top 0.1% rank (100/100) in CS, Class XII Boards 2023 & selected for Reliance UG Scholarship
- AIR 5476 in JEE Advanced 2023 (out of 2.5L+ candidates) & Top 1.42% in JEE Mains 2023 out of 12L+ candidates

Extra-Curriculars

- NSO Volleyball & state/national chess player & EEA frisbee champions